- Higher 2017 sales volumes, incremental year-over-year cost savings associated with restructuring and productivity improvement initiatives, costs associated with various growth investments made in 2016 and changes in currency exchange rates, partially offset by incremental year-over-year costs associated with various product development and sales and marketing growth investments: 60 basis points

Year-over-year operating profit margin comparisons were unfavorably impacted by:

- The incremental year-over-year net dilutive effect of acquired businesses: 20 basis points

2016 COMPARED TO 2015

Year-over-year price increases in the segment contributed 0.3% to sales growth during 2016 as compared to 2015 and are reflected as a component of the change in sales from existing businesses.

Sales from existing businesses in the segment's Transportation Technologies businesses grew at a high-single digit rate during 2016 as compared to 2015, due primarily to strong demand for dispenser, payment and point-of-sale systems, environmental compliance products as well as vehicle and fleet management products, partly offset by weaker year-over-year demand for compressed natural gas products. As expected, beginning in the second half of 2016, the business began to experience reduced EMV-related demand for indoor point-of-sale solutions, as customers had largely upgraded to products that support indoor EMV requirements in the prior year in response to the indoor liability shift. However, demand increased on a year-over-year basis for dispensers and payment systems as customers in the United States continued to upgrade equipment driven primarily by the EMV deadlines related to outdoor payment systems. Geographically, sales from existing businesses continued to increase on a year-over-year basis in the United States and to a lesser extent in Asia and Western Europe.

Sales from existing businesses in the segment's Automation & Specialty Components business declined at a low-single digit rate during 2016 as compared to 2015. The businesses experienced sequential year-over-year improvement in demand during the second half of 2016 as compared to the first half of 2016. During 2016, year-over-year demand declined for engine retarder products due primarily to weakness in the North American heavy-truck market, partly offset by strong growth in China and Europe. In addition, year-over-year demand declined in certain medical and defense related end markets which were partly offset by increased year-over-year demand for industrial automation products particularly in China. Geographically, sales from existing businesses in the segment's Automation & Specialty Components businesses declined in North America, partly offset by growth in Western Europe and China.

Sales from existing businesses in the segment's Franchise Distribution business grew at a mid-single digit rate during 2016, as compared to 2015, due primarily to continued net increases in franchisees as well as continued growth in demand for professional tool products and tool storage products, primarily in the United States. This growth was partly offset by year-over-year declines in wheel service equipment sales during 2016.

Operating profit margins increased 70 basis points during 2016 as compared to 2015. The following factors favorably impacted year-over-year operating profit margin comparisons:

- Higher 2016 sales volumes, pricing improvements, incremental year-over-year cost savings associated with restructuring and productivity improvement initiatives and the incrementally favorable impact of the impairment of certain tradenames used in the segment in 2015 and 2016, net of costs associated with various growth investments, product development and sales and marketing growth investments, higher year-over-year costs associated with restructuring actions and changes in currency exchange rates: 65 basis points

- The incremental net accretive effect in 2016 of acquired businesses: 5 basis points

COST OF SALES AND GROSS PROFIT

	For the Year Ended December 31		
($ in millions)	2017	2016	2015
Sales	$ 6,656.0	$ 6,224.3	$ 6,178.8
Cost of sales	(3,357.5)	(3,191.5)	(3,178.8)
Gross profit	3,298.5	3,032.8	3,000.0
Gross profit margin	49.6%	48.7%	48.6%

The year-over-year increase in cost of sales during 2017 as compared to 2016 is due primarily to the impact of higher year-over-year sales volumes and changes in currency exchange rates partly offset by incremental year-over-year cost savings